

Spin Hamiltonian Framework and Property Evaluation

Summer Internship Report

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Certificate

This is to certify that **Mr. Arif Ahmed Begg**, a second-year student of the *Department of Chemistry, IIT Kharagpur*, bearing Roll No **24CY40005** has successfully completed his internship under my supervision during the period **09/05/2025 to 10/07/2025**. During this time, he has worked on a project titled **"Spin Hamiltonian Framework & Property Evaluation"**.

He demonstrated a strong work ethic, a willingness to learn, and contributed effectively to the tasks and projects assigned. I rate his performance during the internship as Excellent. His internship was an integral part of the academic curriculum of the institute.

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1 Introduction

Spin is an intrinsic form of angular momentum; it plays a critical role in determining the magnetic and quantum mechanical properties of systems. Unlike position or momentum, spin arises naturally from the mathematics of quantum mechanics and is essential in describing the behaviour of quantum particles, with no classical analogue. A spin Hamiltonian is a mathematical operator that describes the energy of a spin system and governs its quantum dynamics. A Spin Hamiltonian can account for various physical interactions, such as coupling between different spins, interactions with external magnetic fields, spin-orbit coupling, and anisotropic effects due to local environments.

This report explores the concept of Spin Hamiltonians from foundational theory to practical applications. We examine how spin operators are defined and combined, how interactions are encoded into Hamiltonians, and how these models are used to interpret experimental data or simulate physical systems

For electrons, $s = \frac{1}{2}$

Total spin angular momentum $|\vec{S}| = \hbar\sqrt{s(s+1)} = \frac{\sqrt{3}}{2}\hbar$

Spin Hamiltonians are important in fields like quantum magnetism and *Electron Paramagnetic Resonance* (EPR). They allow us to model the interactions between spins and external fields. The theoretical foundation of EPR lies in the quantum mechanical treatment of spin, particularly the use of Spin Hamiltonians to describe the energy levels of electron spins in a magnetic field.

The general form of a Spin Hamiltonian may include the following terms:

- **Zeeman interaction:** Interaction of spins with an external magnetic field:

$$\hat{H}_Z = -\gamma\hbar\vec{B} \cdot \vec{S}$$

where γ is the gyromagnetic ratio, $\vec{B}$ is the magnetic field, and $\vec{S}$ is the spin operator.

- **Exchange interaction:** Interaction between multiple spins, typically written as:

$$\begin{aligned}\hat{H}_{\text{exchange}} &= J\vec{S}_1 \cdot \vec{S}_2 \\ &= J(S_1^x S_2^x + S_1^y S_2^y + S_1^z S_2^z)\end{aligned}$$

where J is the *coupling constant*. This models the coupling between two or more spins. It is responsible for *ferromagnetic coupling* and *antiferromagnetic coupling*, depending on the sign of J .

- **Spin-orbit coupling, zero-field splitting or spin-phonon coupling**, depending on the system.

From a computational point of view, Spin Hamiltonians serve as starting points for numerical simulations in quantum mechanics. By constructing matrix representations of these Hamiltonians and diagonalizing them, one can obtain the energies, and other thermodynamic quantities. These simulations allow us to understand systems that are difficult to study analytically, especially when dealing with multiple spin centres. As the number of spins in a quantum system increases, the dimension of its Hilbert space grows exponentially; for N spin-1/2 particles, it becomes 2^N . Their respective spin operators are combined using Tensor (Kronecker) products to build full Hamiltonians in higher dimensional spaces.

In the realm of quantum computation, Spin Hamiltonians acquire an even more critical role. A two-level spin system, such as a spin-1/2 particle, serves as a natural quantum bit or *qubit*, which is the fundamental unit of quantum information. This concept forms the foundation of quantum computation. However, one of the major challenges is to control qubits at room temperature. Researchers are exploring ways to synthesize molecular systems where each unit contains a single unpaired electron, effectively acting as a qubit. The goal is to create a "bed" of such molecules and study how they interact with each other. At room temperature, a key limitation is the short relaxation time of the excited spin state, which causes the quantum information to decay rapidly (Excited spin states relax rapidly due to spin-lattice and spin-spin relaxation, thermal factors, or spin-orbit coupling). If this relaxation can be slowed by minimizing these mechanisms, quantum computing has the potential to revolutionize the world.

In recent years, coupling between spins and other systems, such as phonons, has turned out to be another significant field for research opportunities. Spin-phonon Hamiltonians describe how lattice vibrations can influence the spin states, leading to phenomena like spin relaxation, and so on. These couplings are relevant in solid-state systems, like molecular magnets.

In this report, we explore a range of Spin Hamiltonians, beginning with the basic algebra of spin-1/2 operators and systematically building toward more complex systems involving interactions, external fields, and finite temperatures. Using numerical methods implemented in *Python*, we analyze the behaviour of these systems under different conditions. Key topics include the Heisenberg and Ising models, thermodynamic properties like magnetization and specific heat, and coupled spin-phonon systems. The objective is not only to understand the mathematical structure of these Hamiltonians but also to connect them with physical intuition.

Another great application to these Spin Hamiltonian problems is *Electron Paramagnetic Resonance* (EPR), also known as *Electron Spin Resonance* (ESR), which is a powerful spectroscopic technique used to detect and analyze systems with unpaired electrons. One of the unique strengths of EPR lies in its sensitivity to the local environment of the unpaired electron.

EPR, at its core, is based on the interaction between the magnetic moment of an unpaired electron and an externally applied magnetic field. The electronic level splits up, followed by the splitting due to the nuclear spin, known as *Hyperfine Splitting*. When placed in a magnetic field, the magnetic moment associated with the electron's intrinsic spin aligns either parallel or antiparallel to the field, resulting in discrete energy levels. If microwave radiation of appropriate frequency is applied, it can induce transitions between these energy levels. The absorption of this radiation is what is detected in an EPR experiment. These Hamiltonians can incorporate a wide range of effects, including external magnetic anisotropy, zero-field splitting and so on. By comparing experimental EPR spectra with theoretical predictions from spin Hamiltonians, researchers can extract detailed information about the system under investigation.

The objective of this project is to investigate the quantum mechanical behavior of spin systems through the formulation and simulation of Spin Hamiltonians, with particular emphasis on the evaluation of physical properties via *partition function*. Using the principles of statistical mechanics, the project aims to compute thermodynamic quantities such as average magnetization and heat capacity. A key component involves modeling spin-1/2 and multi-spin systems under the influence of external magnetic fields, including *Zeeman* and *Exchange interactions*. These results are further extended to simulate and interpret EPR spectra, focusing on transition probabilities, magnetic susceptibility, and hyperfine interactions.

2 Theoretical Background

2.1 Basic Theory of Spin

The theory of spin is identical to the general theory of angular momentum. By analogy with the vector angular momentum J , the spin is also represented by the vector operator S , whose components S_x , S_y and S_z obey the same commutation relations as that of the components J .

$$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$$

$$[\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x$$

$$[\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y$$

In addition to this, S^2 and S_z commute; hence they have common eigenvectors:

$$\hat{S}^2|s, m_s\rangle = \hbar^2 s(s+1)|s, m_s\rangle; \quad \hat{S}_z|s, m_s\rangle = \hbar m_s|s, m_s\rangle$$

where, $m_s = -s, -s+1, \dots, s$

Similarly, $\hat{S}_\pm|s, m_s\rangle = \hbar\sqrt{s(s+1) - m_s(m_s \pm 1)}|s, m_s \pm 1\rangle$

where, $\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$

2.2 Spin-1/2 and Pauli Matrices

For $s = 1/2$, $m_s = \pm 1/2$. Thus, we have two states defined by $|\frac{1}{2}, \frac{1}{2}\rangle$ and $|\frac{1}{2}, -\frac{1}{2}\rangle$. The projection of $\vec{S}$ on the z-axis is restricted to two values only: $\pm\frac{\hbar}{2}$, corresponding to spin-up and spin-down. ^[5]

$$\hat{S}^2|s, m_s\rangle = \begin{pmatrix} \langle \frac{1}{2}, \frac{1}{2} | \hat{S}^2 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, \frac{1}{2} | \hat{S}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \\ \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}^2 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}^2 | \frac{1}{2}, -\frac{1}{2} \rangle \end{pmatrix} = \frac{3\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Also, we have:

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The matrix of $\hat{S}_+$ is defined as:

$$\hat{S}_+|s, m_s\rangle = \begin{pmatrix} \langle \frac{1}{2}, \frac{1}{2} | \hat{S}_+ | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, \frac{1}{2} | \hat{S}_+ | \frac{1}{2}, -\frac{1}{2} \rangle \\ \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}_+ | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, -\frac{1}{2} | \hat{S}_+ | \frac{1}{2}, -\frac{1}{2} \rangle \end{pmatrix} = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$\hat{S}_-$ is the conjugate of $\hat{S}_+$. Therefore, these matrices are defined in the following manner:

$$\hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \hat{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Since $\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)$ and $\hat{S}_y = \frac{i}{2}(\hat{S}_- - \hat{S}_+)$, we have

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The joint eigenvectors of $\hat{S}^2$ and $\hat{S}_z$ are expressed in terms of column matrices of two elements, known as *spinors*:

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

When $s = \frac{1}{2}$, we can introduce the *Pauli matrices* σ_x , σ_y and σ_z , which are related to the spin vector as follows:

$$\hat{S} = \frac{\hbar}{2} \sigma$$

Therefore, we have

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

2.3 Partition Function & Property Evaluation

The partition function is central in statistical mechanics. It encodes all thermodynamic information about a system in equilibrium.

For a quantum system (*Canonical Ensemble*):

$$Z = \sum_i e^{-E_i/(k_B T)}$$

where,

- E_i are the energies of the Hamiltonian $\hat{H}$,
- k_B is the Boltzmann constant,
- T is the absolute temperature,
- The sum is over all energy eigenstates i .

For any observable $\hat{A}$, its thermal average is given by:

$$\langle A \rangle = \frac{1}{Z} \sum_i \langle \psi_i | \hat{A} | \psi_i \rangle e^{-E_i/(k_B T)}$$

Examples

$$\text{Average energy: } \langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

$$\text{Magnetization: } \langle M_z \rangle = -\frac{\partial \ln Z}{\partial B}$$

2.4 Tensor Product in Quantum Mechanics

In quantum mechanics, the state of a system is represented by a vector in a Hilbert space $\mathcal{H}$. For two systems, the joint state space is the tensor product: $\mathcal{H}_{\text{total}} = \mathcal{H}_1 \otimes \mathcal{H}_2$

Basis States

If $\{|\uparrow\rangle, |\downarrow\rangle\}$ is a basis for $\mathbb{C}^2$, then a basis for the combined space is given by:

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$$

Operators on Composite Systems

An operator acting on system 1 is extended as $\hat{A} \otimes \hat{I}$, and on system 2 as $\hat{I} \otimes \hat{B}$. Spin-spin interaction is given by:

$$\hat{H} = J \left(\hat{S}_x \otimes \hat{S}_x + \hat{S}_y \otimes \hat{S}_y + \hat{S}_z \otimes \hat{S}_z \right)$$

2.5 Ladder Operators for SHO System

The time-independent Schrödinger equation for a quantum harmonic oscillator is:

$$\frac{p^2}{2m}\psi + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

The Hamiltonian is represented by:

$$H = \frac{1}{2m} [p^2 + (m\omega x)^2]$$

Using ladder operators, the Hamiltonian simplifies to:

$$H = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right) = \hbar\omega \left(aa^\dagger - \frac{1}{2} \right)$$

where $a^\dagger$ is the **creation operator** and a is the **annihilating operator**.

Then,

$$\begin{aligned} a |n\rangle &= \sqrt{n} |n-1\rangle \Rightarrow \langle m | a |n\rangle = \sqrt{n} \delta_{m,n-1} \\ a^\dagger |n\rangle &= \sqrt{n+1} |n+1\rangle \Rightarrow \langle m | a^\dagger |n\rangle = \sqrt{n+1} \delta_{m,n+1} \end{aligned}$$

Writing the operators in matrix form, in the number basis: ^[3]

$$a = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \dots \\ 0 & 0 & \sqrt{2} & 0 & \dots \\ 0 & 0 & 0 & \sqrt{3} & \dots \\ 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad a^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots \\ \sqrt{1} & 0 & 0 & 0 & \dots \\ 0 & \sqrt{2} & 0 & 0 & \dots \\ 0 & 0 & \sqrt{3} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

These matrices act on the basis states $|n\rangle$ to either raise or lower the quantum number n , which we have covered in one of our problems.

2.6 Electron Paramagnetic resonance(EPR)

It is used to detect species with **unpaired electrons** (e.g., radicals, transition metal ions).

Atomic No.	Neutron No.	Spin	Nature
Even	Even	0	Inactive
Odd	Even	$1/2, 3/2, 5/2 \dots$	Active
Even	Odd	$1/2, 3/2, 5/2 \dots$	Active
Odd	Odd	$1, 2, 3 \dots$	Active

Hamiltonian of the System

The EPR Hamiltonian is given by: ^[6]

$$\hat{H} = \underbrace{\mu_B \mathbf{B} \cdot \mathbf{g} \cdot \hat{\mathbf{S}}}_{ZeemanTerm} + \underbrace{\hat{\mathbf{S}} \cdot \mathbf{A} \cdot \hat{\mathbf{I}}}_{HyperfineTerm}$$

- **Zeeman Splitting:** Interaction of magnetic field with electron spin.
- **Hyperfine Splitting:** Interaction of electron spin with nuclear spin.

Zeeman Splitting

In a magnetic field applied along z -axis, the M_s levels split into its respective levels (Degeneracy is lost). When the energy from the externally applied magnetic field matches to this energy gap, a transition is observed. The Hamiltonian and the energy levels are given by the following expressions:

$$\hat{H}_{Zeeman} = \mu_B g B_z \hat{S}_z \Rightarrow E_{\pm} = \pm \frac{1}{2} g \mu_B B$$

Resonance occurs when the energy gap matches the microwave frequency of the spectrometer: ^[1]

$$h\nu = g\mu_B B \Rightarrow \nu = \frac{g\mu_B B}{h}$$

In EPR experiments, the microwave frequency ν is fixed and the magnetic field B is swept to satisfy the resonance condition.

Hyperfine Splitting^[4]

Hyperfine splitting arises due to the nuclear spin, which interacts with the electronic magnetic moment and splits the electronic Zeeman levels, and its Hamiltonian is given by:

$$\hat{H}_{Hyperfine} = \hat{\mathbf{S}} \cdot \mathbf{A} \cdot \hat{\mathbf{I}}$$

Selection Rules

$$\Delta m_S = \pm 1, \quad \Delta m_I = 0$$

Example: Hydrogen Radical (H·)

- Electron spin: $S = \frac{1}{2} \Rightarrow m_S = \pm \frac{1}{2}$
- Nuclear spin: $I = \frac{1}{2} \Rightarrow m_I = \pm \frac{1}{2}$

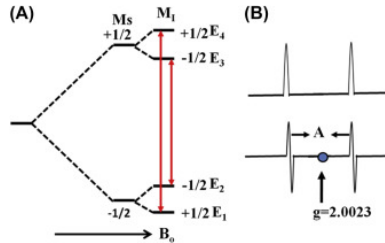


Figure 1: EPR splitting pattern for a H radical

Anisotropy in EPR

We have so far assumed that the Zeeman interaction depends only on the angle between the effective spin vector and the external magnetic field. However, in practice, this interaction can also depend on the angle that the magnetic field $\vec{B}$ makes with specific molecular axes defined by the local symmetry of the paramagnetic complex. This directional dependence is known as **g-anisotropy** [2]. To account for this anisotropy, the Zeeman Hamiltonian is expressed in terms of the **g-tensor**. The general form of the Zeeman Hamiltonian becomes:

$$\mathcal{H} = \beta(\mathbf{H} \cdot \mathbf{g} \cdot \tilde{\mathbf{S}})$$

which is a shorthand notation for

$$\begin{aligned} \mathcal{H} = \beta \{ & g_{xx} H_x \tilde{S}_x + g_{yy} H_y \tilde{S}_y + g_{zz} H_z \tilde{S}_z + g_{xy} H_x \tilde{S}_y \\ & + g_{yx} H_y \tilde{S}_x + g_{yz} H_y \tilde{S}_z + g_{zy} H_z \tilde{S}_y + g_{zx} H_z \tilde{S}_x + g_{xz} H_x \tilde{S}_z \} \end{aligned}$$

where:

- μ_B is the *Bohr Magneton*,
- $\vec{B}$ is the magnetic field vector,
- $\mathbf{g}$ is the 3×3 g-tensor,
- $\hat{\vec{S}}$ is the electron spin operator vector.

If the g-tensor is diagonalised in its *principal axes frame* (no off-diagonal terms), the Hamiltonian simplifies to:

$$\hat{H}_Z = \mu_B \left(g_x B_x \hat{S}_x + g_y B_y \hat{S}_y + g_z B_z \hat{S}_z \right)$$

The $\mathbf{g}$ -tensor in an anisotropic field is given by:

$$\mathbf{g} = \begin{pmatrix} g_{xx} & g_{xy} & g_{xz} \\ g_{yx} & g_{yy} & g_{yz} \\ g_{zx} & g_{zy} & g_{zz} \end{pmatrix}$$

Simplifying this under principal axes, we get:

$$\mathbf{g} = \begin{pmatrix} g_{xx} & 0 & 0 \\ 0 & g_{yy} & 0 \\ 0 & 0 & g_{zz} \end{pmatrix}$$

Our upcoming project aims to implement this anisotropy effect in simple organic radicals, and study their EPR spectra in *ORCA*.

3 Results & Discussions

We have solved quite a lot *Spin Hamiltonians* problems. Here, we shall present a few of them. We have also simulated a simple EPR spectrum, which is a simple application of these *Spin Hamiltonians*. We shall present the *Python* codes and try to interpret the results.

3.1 Spin Algebra and Representations

3.1.1 Creating the Pauli Matrices for an Arbitrary Spin

Python code snippet:

```
def print_spin_operators(S):
    hbar= 1.0
    d = int(2 * S + 1) #Dimension of the matrix
    m = np.linspace(S, -S, d)
    print("m values:", m)
    S_plus = np.zeros((d, d), dtype=complex)
    Sz = np.zeros((d, d), dtype=complex)
    for i in range(d):
        Sz[i, i] = m[i] * hbar
    for i in range(d - 1):
        S_plus[i, i + 1] = hbar * np.sqrt(S * (S + 1) - m[i] * (m[i + 1]))
    S_minus = S_plus.T.conj()
    Sx = 0.5 * (S_plus + S_minus)
    Sy = -0.5j * (S_plus - S_minus)
    S2 = S * (S + 1) * np.eye(d)
    #Print the operators now
print_spin_operators(1) #Creating Pauli Matrices corresponding to spin 1
```

Output:

```
m values: [ 1.  0. -1.]
Spin S = 1
Sx =
[[0.          +0.j 0.70710678+0.j 0.          +0.j]
 [0.70710678+0.j 0.          +0.j 0.70710678+0.j]
 [0.          +0.j 0.70710678+0.j 0.          +0.j]]
Sy =
[[0.-0.j          0.-0.70710678j 0.-0.j          ]
 [0.+0.70710678j 0.-0.j          0.-0.70710678j]
 [0.-0.j          0.+0.70710678j 0.-0.j          ]]
Sz =
[[ 1.+0.j  0.+0.j  0.+0.j]
 [ 0.+0.j  0.+0.j  0.+0.j]
 [ 0.+0.j  0.+0.j -1.+0.j]]
S^2 =
[[2.  0.  0.]
 [0.  2.  0.]
 [0.  0.  2.]]
```

Figure 2: Output to 3.1.1

3.1.2 Total Spin of Two Particles

The Hamiltonian to this system is represented by : $S_{tot}^2 = (\vec{S}_1 + \vec{S}_2)^2$

Python code snippet:

```
singlet = triplet = 0

S1x = np.kron(Sx, I)
S1y = np.kron(Sy, I)
S1z = np.kron(Sz, I)
S2x = np.kron(I, Sx)
S2y = np.kron(I, Sy)
S2z = np.kron(I, Sz)

S_tot2 = (S1x + S2x) @ (S1x + S2x) + (S1y + S2y) @ (S1y + S2y)
        \ + (S1z + S2z) @ (S1z + S2z)
print(S_tot2)
eigvals, eigvecs = np.linalg.eigh(S_tot2)

for i in range(len(eigvals)):
    if eigvals[i] == 0.0:
        singlet+=1
    else:
        triplet+=1

Sz_tot = Sz
Sz_evals, Sz_evecs = np.linalg.eigh(Sz_tot)
```

Output:

```
[[2.+0.j 0.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 1.+0.j 1.+0.j 0.+0.j]
 [0.+0.j 1.+0.j 1.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 0.+0.j 2.+0.j]]
```

Eigenvalues of S^2 for Spin of Two Particles (in units of $\hbar^2$) : [0. 2. 2. 2.]

Eigenvalues of S_z (in units of $\hbar$): [-0.5 0.5]

Number of singlet states: 1

Number of triplet states: 3

Figure 3: Output to 3.1.2

Interpretation:

When considering a system of two spin- $\frac{1}{2}$ particles, the Hilbert space is of dimension $2 \times 2 = 4$. The total spin operator $\hat{S}_{\text{total}}^2$ now has eigenvalues: $[0, 2, 2, 2]$ (in units of $\hbar^2$)

These values correspond to:

- One eigenvalue 0: corresponds to the **singlet state** with total spin $S = 0$
- Three eigenvalues 2: each corresponds to a **triplet state** with total spin $S = 1$, since $S(S+1)\hbar^2 = 2\hbar^2$ when $S = 1$

The spin projection operator $\hat{S}_z$ has eigenvalues: $[-0.5, 0.5]$ (in units of $\hbar$)

Hence, the system consists of one singlet state and three triplet states, consistent with spin addition rules for two spin- $\frac{1}{2}$ particles.

3.2 Spin Hamiltonians and Energetics

3.2.1 Heisenberg Hamiltonian

$$H = J\vec{S}_1 \cdot \vec{S}_2$$

where, S_1 and S_2 represent the spins of electrons 1 and 2, respectively.

Python code snippet:

```
S1x = np.kron(Sx, I) #Creating the Hilbert Space
S1y = np.kron(Sy, I)
S1z = np.kron(Sz, I)
S2x = np.kron(I, Sx)
S2y = np.kron(I, Sy)
S2z = np.kron(I, Sz)

basis = ["|↑↑⟩", "|↑↓⟩", "|↓↑⟩", "|↓↓⟩"]
styles = ['-', '--', ':', '-.']

def heisenberg_hamiltonian(J):
    return J * (S1x @ S2x + S1y @ S2y + S1z @ S2z)
print("\n===== HEISENBERG MODEL =====")
J_vals = np.linspace(-1, 1, 5)
energy_levels = []

for j in J_vals:
    H = heisenberg_hamiltonian(j)
    eigvals, eigvecs = np.linalg.eigh(H)
    energy_levels.append(eigvals)
    if j in (J_vals[0], J_vals[-1], 0):
        print(f"\nJ = {j:.2f}")

for i in range(4):
    vec = eigvecs[:, i]
```

```

if j in (J_vals[0], J_vals[-1], 0):
    print(f"Energy Level {i+1}: {eigvals[i]:.3f} ;", end="")
    print(" Eigenvector : ", end="")
    l=""
    for (amp, state) in zip(vec, basis):
        if abs(amp) > 0.0001:
            l= l+ f" {np.real(amp):+0.3f} {state}"
    print(l)
energy_levels = np.array(energy_levels)
# Plot the levels now, as usual

```

Output:

```

===== HEISENBERG MODEL =====

J = -1.00
Energy Level 1: -0.250 ; Eigenvector : +1.000 |↑↑)
Energy Level 2: -0.250 ; Eigenvector : -0.707 |↑↓) -0.707 |↓↑)
Energy Level 3: -0.250 ; Eigenvector : +1.000 |↓↓)
Energy Level 4: 0.750 ; Eigenvector : -0.707 |↑↑) +0.707 |↓↓)

J = 0.00
Energy Level 1: 0.000 ; Eigenvector : +1.000 |↑↑)
Energy Level 2: -0.000 ; Eigenvector : +1.000 |↑↓)
Energy Level 3: -0.000 ; Eigenvector : +1.000 |↓↑)
Energy Level 4: 0.000 ; Eigenvector : +1.000 |↓↓)

J = 1.00
Energy Level 1: -0.750 ; Eigenvector : +0.707 |↑↓) -0.707 |↓↑)
Energy Level 2: 0.250 ; Eigenvector : +1.000 |↑↑)
Energy Level 3: 0.250 ; Eigenvector : +0.707 |↑↑) +0.707 |↓↑)
Energy Level 4: 0.250 ; Eigenvector : +1.000 |↓↓)

Eigenvalues summary :
[[-0.25 -0.25 -0.25  0.75]
 [-0.125 -0.125 -0.125  0.375]
 [ 0.    -0.    -0.    0.   ]
 [-0.375  0.125  0.125  0.125]
 [-0.75   0.25   0.25   0.25]]

```

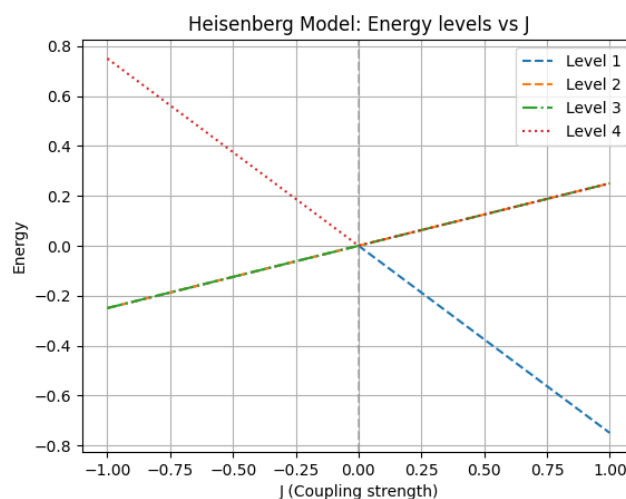


Figure 4: Output to 3.2.1

Interpretation:

We consider the Heisenberg Hamiltonian for two interacting spin- $\frac{1}{2}$ particles:

$$\hat{H} = J \hat{\vec{S}}_1 \cdot \hat{\vec{S}}_2 = J \left(\hat{S}_{1x} \hat{S}_{2x} + \hat{S}_{1y} \hat{S}_{2y} + \hat{S}_{1z} \hat{S}_{2z} \right)$$

The eigenstates of the total spin system are:

Triplet States ($S = 1$, Symmetric)

$$\begin{aligned} |T_+\rangle &= |\uparrow\uparrow\rangle \\ |T_0\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ |T_-\rangle &= |\downarrow\downarrow\rangle \end{aligned}$$

Singlet State ($S = 0$, Antisymmetric)

$$|S_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Eigenvalues

The eigenvalues of the Hamiltonian depend on the total spin S :

- Triplet states ($S = 1$): $E = \frac{J}{4}$
- Singlet state ($S = 0$): $E = -\frac{3J}{4}$

At negative values of J , the triplet states become energetically favourable due to ferromagnetic coupling, which promotes parallel spin alignment. Conversely, for positive J , the singlet state is favoured, indicating antiferromagnetic coupling that favours antiparallel spin configuration.

3.2.2 Ising Hamiltonian in Magnetic Field

$$H = JS_{1z}S_{2z} - h(S_{1z} + S_{2z}), \text{ where } h \text{ is the external magnetic field}$$

Python code snippet:

```
def ising_hamiltonian(H):
    return -J * (S1z @ S2z) - H * (S1z + S2z)
# Tested for J=3(AFC). The rest is similar to the Heisenberg Hamiltonian.
```

Output:

```

===== ISING MODEL =====

H = -1.00
Energy Level 1: -1.750 ; Eigenvector: +1.000 |↓↓⟩
Energy Level 2: 0.250 ; Eigenvector: +1.000 |↑↑⟩
Energy Level 3: 0.750 ; Eigenvector: +1.000 |↓↑⟩
Energy Level 4: 0.750 ; Eigenvector: +1.000 |↑↓⟩

H = 0.00
Energy Level 1: -0.750 ; Eigenvector: +1.000 |↑↑⟩
Energy Level 2: -0.750 ; Eigenvector: +1.000 |↓↓⟩
Energy Level 3: 0.750 ; Eigenvector: +1.000 |↓↑⟩
Energy Level 4: 0.750 ; Eigenvector: +1.000 |↑↓⟩

H = 1.00
Energy Level 1: -1.750 ; Eigenvector: +1.000 |↑↑⟩
Energy Level 2: 0.250 ; Eigenvector: +1.000 |↓↓⟩
Energy Level 3: 0.750 ; Eigenvector: +1.000 |↓↑⟩
Energy Level 4: 0.750 ; Eigenvector: +1.000 |↑↓⟩

Eigenvalues summary :
[[-1.75 0.25 0.75 0.75]
 [-1.25 -0.25 0.75 0.75]
 [-0.75 -0.75 0.75 0.75]
 [-1.25 -0.25 0.75 0.75]
 [-1.75 0.25 0.75 0.75]]
    
```

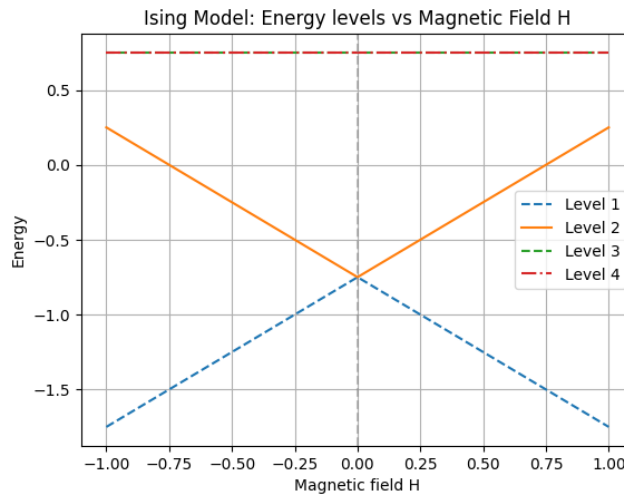


Figure 5: Output to 3.2.2

Interpretation:

We consider the Ising Hamiltonian for two interacting spin- $\frac{1}{2}$ particles in an external magnetic field :

$$\hat{H} = J \hat{S}_{1z} \hat{S}_{2z} - h(\hat{S}_{1z} + \hat{S}_{2z})$$

We work in the standard basis : $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$.

As the external magnetic field, h varies from negative to positive, the ground state switches between $|\downarrow\downarrow\rangle$ and $|\uparrow\uparrow\rangle$, reflecting the competition between the magnetic field and spin interaction.

3.3 Evaluation of Thermodynamic Properties

3.3.1 Partition Function and Magnetization

$$H = J \vec{S}_1 \cdot \vec{S}_2 + \mu B (S_{1z} + S_{2z})$$

$$Z = \sum_i e^{-E_i/(k_B T)}$$

$$\langle M_z \rangle = \frac{1}{Z} \sum_i \langle \psi_i | S_{1z} + S_{2z} | \psi_i \rangle e^{-E_i/(k_B T)}$$

Python code snippet:

```
S1z = np.kron(Sz, I)
S2z = np.kron(I, Sz)
H0 = np.kron(Sx, Sx) + np.kron(Sy, Sy) + np.kron(Sz, Sz)

def magnetization(eigenvalues, eigenvectors):
    Mz_op = S1z + S2z
    w = np.exp(-eigenvalues / (kB * T)) # pf components
    Z = np.sum(w) # Partition function calculation
    mz_values = (eigenvectors.T @ Mz_op @ eigenvectors)
    return np.sum(mz_values * w) / Z

Mz_val = []
for B in B_val:
    H = J * H0 + mu * B * (S1z + S2z)
    eigenvalues, eigenvectors = np.linalg.eigh(H) # Eigenvalues
    Mz_val.append(magnetization(eigenvalues, eigenvectors))
#Now plot <Mz> vs B as usual
```

Output:

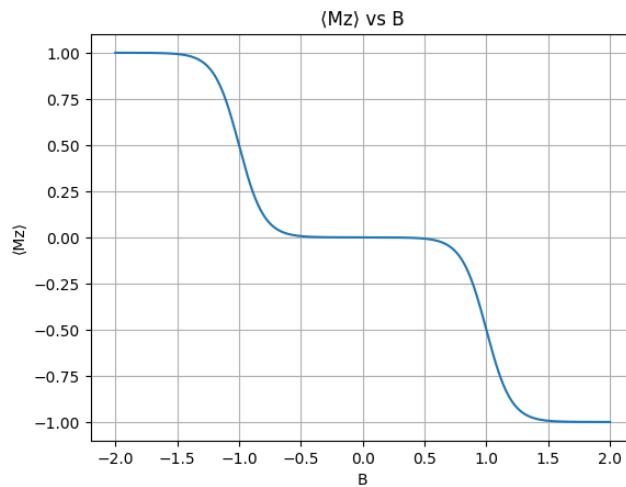


Figure 6: Plot of $\langle M_z \rangle$ vs B , $J = 1$ and $T = 0.1$

Interpretation:

For $J = 1$, the Heisenberg interaction between the two spin- $\frac{1}{2}$ particles is antiferromagnetic. At $B = 0$, the system favors the **singlet state**, which has total spin zero and thus $\langle M_z \rangle = 0$.

As the external magnetic field increases, the **triplet states**, which have total spin 1 and magnetic quantum numbers $M_z = +1, 0, -1$ become energetically favorable due to the Zeeman term. At large positive or negative B , the spins align with the field, resulting in $\langle M_z \rangle \approx \pm 1$.

3.3.2 Specific Heat and Magnetic Susceptibility

$$1. \text{ Internal energy: } \langle E \rangle = \frac{1}{Z} \sum_i E_i e^{-E_i/(k_B T)}$$

$$2. \text{ Specific heat: } C_v = \frac{1}{k_B T^2} (\langle E^2 \rangle - \langle E \rangle^2)$$

$$\text{where } \langle E^2 \rangle = \frac{1}{Z} \sum_i E_i^2 e^{-E_i/(k_B T)}$$

$$3. \text{ Magnetic susceptibility: } \chi = \frac{1}{k_B T} (\langle M_z^2 \rangle - \langle M_z \rangle^2)$$

$$\text{where } \langle M_z^2 \rangle = \frac{1}{Z} \sum_i (\psi_i | S_{1z} + S_{2z} | \psi_i)^2 e^{-E_i/(k_B T)}$$

Python code snippet:

```
S1z = np.kron(Sz, I)
S2z = np.kron(I, Sz)
H0 = np.kron(Sx, Sx) + np.kron(Sy, Sy) + np.kron(Sz, Sz)

def sp_heat(eigenvalues, T):
    w = np.exp(-eigenvalues / (kB * T))
    Z = np.sum(w) # Z = P.F
    E_avg = np.sum(eigenvalues * w) / Z
    E2_avg = np.sum((eigenvalues**2) * w) / Z
    return ((E2_avg - E_avg**2) / (kB * T**2))

def susceptibility(eigenvalues, eigenvectors, T):
    Mz_op = S1z + S2z
    w = np.exp(-eigenvalues / (kB * T))
    Z = np.sum(w)
    mz_values = (eigenvectors.T @ Mz_op @ eigenvectors)
    Mz_avg = np.sum(mz_values * w) / Z
    Mz2_avg = np.sum((mz_values**2) * w) / Z
    return ((Mz2_avg - Mz_avg**2) / (kB * T))

plt.figure(figsize=(8, 4))
Cv_vals = []
```

```
Chi_vals = []

for T in T_values:
    H = J * H0 + mu * B * (S1z + S2z) # Hamiltonian
    eigenvalues, eigenvectors = np.linalg.eigh(H)
    Cv_vals.append(sp_heat(eigenvalues, T))
    Chi_vals.append(susceptibility(eigenvalues, eigenvectors, T))
```

Output:

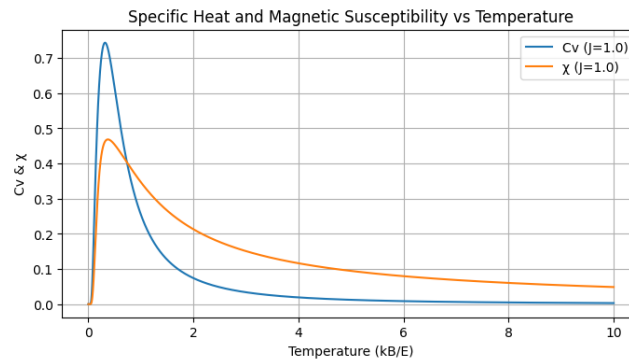


Figure 7: Plot of C_v and χ vs T for $J=1$ and $B=0.5$

Interpretation:

The plot shows the variation of the specific heat C_v and magnetic susceptibility χ with temperature for a two-spin system with exchange interaction strength $J = 1.0$.

Specific Heat C_v

- The specific heat exhibits a sharp peak at a low temperature.
- This peak corresponds to the temperature range, where thermal energy is sufficient to excite the system from the ground state to higher spin states. Beyond the peak, C_v decreases rapidly and approaches zero.

Magnetic Susceptibility χ

- At low temperature, the system's response to magnetic field is strongest when spins can align parallelly.
- As temperature increases, thermal fluctuations dominate, disrupting spin alignment and causing a gradual decrease in χ , which eventually approaches zero asymptotically.

3.4 Spin-Phonon Hamiltonians

3.4.1 Quantum Harmonic Oscillators

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

$\hat{a}^\dagger$ represents raising operator and $\hat{a}$ represents the lowering operator for Harmonic Oscillators.

Python code snippet:

```
N = 5
hbar = 1.0
omega = 1.0

a = np.zeros((N, N))

for i in range(N-1):
    a[i, i+1] = np.sqrt(i+1)

adag = a.T

H = hbar * omega * (adag @ a + 0.5 * np.eye(N))
print("Hamiltonian matrix:\n", H)

eigenvalues, eigenvectors = np.linalg.eigh(H)
print("\nEnergy levels: ", eigenvalues)
```

Output:

```
Hamiltonian matrix:
[[0.5 0.  0.  0.  0. ]
 [0.  1.5 0.  0.  0. ]
 [0.  0.  2.5 0.  0. ]
 [0.  0.  0.  3.5 0. ]
 [0.  0.  0.  0.  4.5]]

Energy levels:  [0.5 1.5 2.5 3.5 4.5]
```

Figure 8: Output to 3.4.1

Interpretation:

The corresponding energy levels are:

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega, \quad n = 0, 1, 2, 3, 4$$

This is the spectrum of the quantum harmonic oscillator, where $\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + \frac{1}{2})$. The system has equally spaced energy levels, each differing by $\hbar\omega$, with a ground state energy of $\hbar\omega/2$.

The ladder operators satisfy the relations:

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad \hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

These operators allow transitions between adjacent energy levels, forming the foundation of quantum oscillator dynamics.

3.4.2 Coupled Spin-Phonon Systems

$$\hat{H} = \frac{1}{2} \hbar \omega_0 \sigma_z + \hbar \omega \hat{a}^\dagger \hat{a} + g \sigma_z (\hat{a} + \hat{a}^\dagger)$$

Python code snippet:

```
omega = 1.0
omega0 = 1.0
g = 1
N = 2
I_spin = np.eye(2, dtype=complex)

def phonon():
    a = np.zeros((N, N), dtype=complex)
    for n in range(1, N):
        a[n-1, n] = np.sqrt(n)
    return a

a = phonon()
adag = a.T.conj()
I_phonon = np.eye(N, dtype=complex)

H_spin = 0.5 * hbar * omega0 * np.kron(sigz, I_phonon)
H_phonon = hbar * omega * np.kron(I_spin, adag @ a)
H_int = g * np.kron(sigz, (adag + a))
H = H_spin + H_phonon + H_int
print(np.real(H))
eigvals, eigvecs = np.linalg.eigh(H)
print(eigvals)
```

Output:

```
[[ 0.5  1.   0.   0. ]
 [ 1.   1.5  0.   0. ]
 [ 0.   0.  -0.5 -1. ]
 [ 0.   0.  -1.   0.5]]
[-1.11803399 -0.11803399  1.11803399  2.11803399]
```

Figure 9: Output to 3.4.2

Interpretation:

We consider a system where a spin- $\frac{1}{2}$ particle is coupled to a single-mode phonon field. The Hamiltonian for the uncoupled system is:

$$\hat{H} = \frac{1}{2}\hbar\omega_0 \sigma_z \otimes I_{\text{ph}} + \hbar\omega I_{\text{spin}} \otimes \hat{a}^\dagger \hat{a}$$

where:

- σ_z is the Pauli matrix representing the spin's energy levels,
- ω_0 is the spin transition frequency,
- ω is the phonon frequency.

The total Hilbert space is a tensor product of the spin and phonon spaces: $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{spin}} \otimes \mathcal{H}_{\text{phonon}}$. The eigenvalues for spin up states are:

$$E_n^{(\uparrow)} = \frac{1}{2}\hbar\omega_0 + n\hbar\omega$$

and for spin down:

$$E_n^{(\downarrow)} = -\frac{1}{2}\hbar\omega_0 + n\hbar\omega$$

For $N = 2$ phonon levels, the full Hamiltonian matrix is of size 4×4 , which upon diagonalisation yield the eigenvalues. These values show deviations from perfect harmonic spacing, *indicating coupling between spin and phonon states*.

Analytically, one can create the Hamiltonian in the following fashion:

$\hat{H}$	$ \frac{1}{2}, 0\rangle$	$ \frac{1}{2}, 1\rangle$	$ \frac{-1}{2}, 0\rangle$	$ \frac{-1}{2}, 1\rangle$
$\langle\frac{1}{2}, 0 $	$\frac{\hbar\omega_0}{2}$	g	0	0
$\langle\frac{1}{2}, 1 $	g	$\frac{\hbar\omega_0}{2} + \hbar\omega$	0	0
$\langle\frac{-1}{2}, 0 $	0	0	$-\frac{\hbar\omega_0}{2}$	g
$\langle\frac{-1}{2}, 1 $	0	0	g	$-\frac{\hbar\omega_0}{2} + \hbar\omega$

Table 1: Matrix elements of the Hamiltonian $\hat{H}$ in the basis $|s, n\rangle$

Now, if $\hbar, \omega$ and ω_0 are considered to be unity in atomic units, then one can write the Hamiltonian in this matrix fashion:

$$\hat{H} = \begin{pmatrix} \frac{1}{2} & g & 0 & 0 \\ g & \frac{3}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & -g \\ 0 & 0 & -g & \frac{1}{2} \end{pmatrix}$$

3.5 Simple EPR Spectra Simulator

This simulation models the EPR spectrum of an electron with spin $S = \frac{1}{2}$, coupled to a *single nucleus* via the so-called isotropic hyperfine interaction. The user specifies the nuclear spin I and the hyperfine constant A (in Tesla). The Hamiltonian includes the electron Zeeman term and hyperfine coupling. Transition intensities under a 9 GHz microwave field are calculated to generate the absorption and derivative spectra. It is based on the Hamiltonian that we have discussed in the *Theoretical Background* section.

Python code snippet:

```
h = 6.626e-34
mu_B = 9.27e-24
g_e = 2.0023 #Isotropic

I = float(input("Enter spin of nucleus: "))
A_T = float(input("Enter hyperfine constant (Tesla): "))
freq_mw_Hz = 9e9

A_Hz = g_e * mu_B * A_T / h
Ix, Iy, Iz = spin_operators(I)
Ie = np.eye(2)
In = np.eye(int(2 * I + 1))

# Total operators- Creating the Hilbert space
Sx_tot = np.kron(Sx, In)
Sy_tot = np.kron(Sy, In)
Sz_tot = np.kron(Sz, In)
Ix_tot = np.kron(Ie, Ix)
Iy_tot = np.kron(Ie, Iy)
Iz_tot = np.kron(Ie, Iz)

B_field = np.linspace(0.20, 0.40, 500)
width = 150e6 #To adjust the width of the plot
spectrum = np.zeros(len(B_field))

for i, B in enumerate(B_field):
    H_zeeman = g_e * mu_B * B / h * Sz_tot
    H_hyperfine = A_Hz * (Sx_tot @ Ix_tot + Sy_tot @ Iy_tot + Sz_tot @ Iz_tot)
    H = H_zeeman + H_hyperfine
    energies, states = np.linalg.eigh(H)
    intensity_sum = 0
    for m in range(len(energies)):
        for n in range(m + 1, len(energies)):
            delta_f = energies[n] - energies[m] # in Hz
            transition_element = states[:, n].conj().T @ Sx_tot @ states[:, m]
            intensity = np.abs(transition_element)**2
            w = np.exp(-((delta_f - freq_mw_Hz)/width)**2)
            intensity_sum += intensity * w
```

```

spectrum[i] = intensity_sum
spectrum /= np.max(spectrum) #Normalizing the spectra
d_spectrum_dB = np.gradient(spectrum, B_field)
d_spectrum_dB /= np.max(np.abs(d_spectrum_dB))
#Plot the Absorption & the Derivative plot as usual

```

Output:

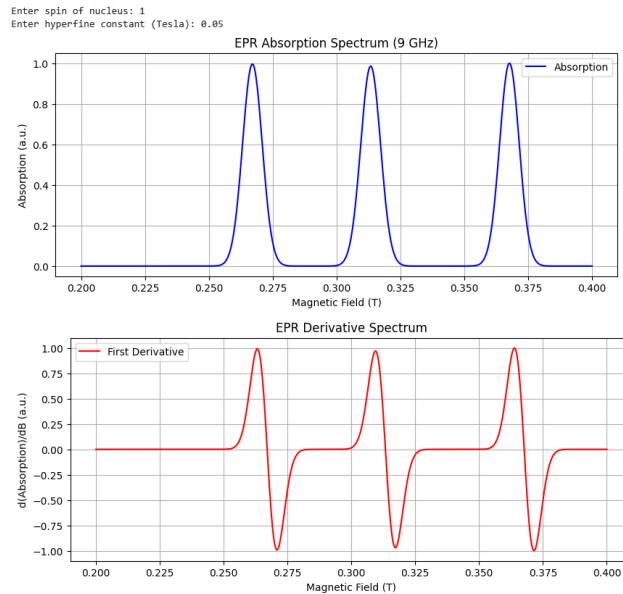


Figure 10: Simulated EPR spectra for $I=1$ and $A=0.05$ T

Interpretation:

For a nucleus with spin $I = 1$, the hyperfine interaction will result in splitting the electron spin levels into $2I + 1 = 3$ components due to *coupling between the electron spin $S = \frac{1}{2}$ and the nuclear spin*.

As a result, the EPR absorption spectrum (top panel) exhibits three distinct peaks, each satisfying the selection rules of EPR. The peak separation is directly proportional to the hyperfine constant A , which here is taken as 0.05 T, i.e the separation between two neighbouring peaks is the hyperfine constant. The bottom panel shows the first derivative of the absorption spectrum, which corresponds to the inflection points of the absorption peaks and reflects the spectral transitions more clearly.

This spectrum confirms the expected quantum mechanical behaviour of an electron coupled isotropically to a *spin-1* nucleus.

4 Conclusion & Future Outlook

Conclusion

In this project, we studied the quantum behaviour of spin systems. Spin Hamiltonians such as the Heisenberg and Ising models were explored in detail. Starting from simple two-spin systems, we extended our analysis to three- and four-spin chains, incorporating both open and periodic boundary conditions. We also expanded our idea to the thermal properties of spin systems, including the specific heat and magnetic susceptibility.

We also implemented a spin–phonon coupled Hamiltonian and observed the effect of vibrational degrees of freedom on the energy spectrum.

Future Outlook

- **Advanced EPR Simulation in ORCA:** Future work will involve performing high level EPR calculations on organic radicals using ORCA, including anisotropic interactions. Spectral simulations can be directly compared with experimental EPR data.
- **Low-Temperature EPR Dynamics:** We will also be exploring EPR transitions at low temperatures, where one can observe suppressed spin lattice interactions, computationally. Special emphasis will be given to studying how to minimize decoherence pathways.
- **Spin Relaxation Modelling:** By incorporating spin–phonon coupling or external noise, we can estimate spin relaxation times (T_1 , T_2) and study decoherence mechanisms relevant to quantum information science. The techniques developed here can be extended to study *spin qubits*.
- **Spin–Phonon Coupling and Decoherence:** Future work can include a detailed study of how spin states interact with phonons, which play a major role in relaxation and decoherence. Modelling this interaction can help understand and control spin dynamics in solid-state systems.

References

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